Civil Engineering

Trigonometry, Bearings and Complex Numbers

Professional Learner Notes

Trigonometric ratios, Pythagoras, sine rule, cosine rule, bearings and introductory complex numbers.

Section	Purpose
Right-triangle trigonometry	Sine, cosine and tangent in construction geometry.
Angles of elevation and depression	Heights, slopes and inaccessible distances.
Sine and cosine rules	Non-right-angled triangles in surveying and layout.
Bearings	Whole-circle bearings and direction calculation.
Complex numbers	Rectangular and polar forms for advanced engineering mathematics.

How to Study This Topic

The learning method follows a programmed engineering mathematics approach: understand the concept, study the formula, follow a worked method, then practise with a technical problem. The aim is not memorisation only; the learner must interpret the answer in the department context.

- Write the formula before substitution.
- Use correct units.
- Check whether the result is reasonable.
- State a technical interpretation.
- Repeat practice until the method is fluent.

Formula Summary

Pythagoras

$$a^2+b^2=c^2$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Sine

$$\sin\theta=\frac{O}{H}$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Cosine rule

$$c^2=a^2+b^2-2ab\cos C$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Sine rule

$$\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Complex form

$$z=a+bi$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

1. Right-triangle trigonometry

Sine, cosine and tangent in construction geometry.

In civil engineering, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

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a^2+b^2=c^2
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Worked method

Example: Height of retaining wall. From a point 12 m away, the angle of elevation to the top of a wall is 32°. Estimate the height.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports heights. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving triangulation. Show formula, substitution, units and interpretation.

2. Create and solve one technical calculation involving bearings. Show formula, substitution, units and interpretation.

3. Create and solve one technical calculation involving polar representation. Show formula, substitution, units and interpretation.

4. Create and solve one technical calculation involving heights. Show formula, substitution, units and interpretation.

5. Create and solve one technical calculation involving slopes. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving triangulation. Show formula, substitution, units and interpretation.

7. Create and solve one technical calculation involving bearings. Show formula, substitution, units and interpretation.

8. Create and solve one technical calculation involving polar representation. Show formula, substitution, units and interpretation.

2. Angles of elevation and depression

Heights, slopes and inaccessible distances.

In civil engineering, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

```
\sin\theta=\frac{O}{H}
```

Worked method

Example: Road reserve triangle. Given two sides and an included angle, use the cosine rule to find a missing boundary length.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports slopes. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving bearings. Show formula, substitution, units and interpretation.

2. Create and solve one technical calculation involving polar representation. Show formula, substitution, units and interpretation.

3. Create and solve one technical calculation involving heights. Show formula, substitution, units and interpretation.

4. Create and solve one technical calculation involving slopes. Show formula, substitution, units and interpretation.

5. Create and solve one technical calculation involving triangulation. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving bearings. Show formula, substitution, units and interpretation.

7. Create and solve one technical calculation involving polar representation. Show formula, substitution, units and interpretation.

8. Create and solve one technical calculation involving heights. Show formula, substitution, units and interpretation.

3. Sine and cosine rules

Non-right-angled triangles in surveying and layout.

In civil engineering, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$c^2=a^2+b^2-2ab\cos C$$

Worked method

Example: Bearing computation. A line has departure 45 m east and latitude 70 m north. Estimate its bearing.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports triangulation. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving polar representation. Show formula, substitution, units and interpretation.

2. Create and solve one technical calculation involving heights. Show formula, substitution, units and interpretation.

3. Create and solve one technical calculation involving slopes. Show formula, substitution, units and interpretation.

4. Create and solve one technical calculation involving triangulation. Show formula, substitution, units and interpretation.

5. Create and solve one technical calculation involving bearings. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving polar representation. Show formula, substitution, units and interpretation.

7. Create and solve one technical calculation involving heights. Show formula, substitution, units and interpretation.

8. Create and solve one technical calculation involving slopes. Show formula, substitution, units and interpretation.

4. Bearings

Whole-circle bearings and direction calculation.

In civil engineering, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

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\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}
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Worked method

Example: Height of retaining wall. From a point 12 m away, the angle of elevation to the top of a wall is 32°. Estimate the height.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports bearings. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving heights. Show formula, substitution, units and interpretation.

2. Create and solve one technical calculation involving slopes. Show formula, substitution, units and interpretation.

3. Create and solve one technical calculation involving triangulation. Show formula, substitution, units and interpretation.

4. Create and solve one technical calculation involving bearings. Show formula, substitution, units and interpretation.

5. Create and solve one technical calculation involving polar representation. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving heights. Show formula, substitution, units and interpretation.

7. Create and solve one technical calculation involving slopes. Show formula, substitution, units and interpretation.

8. Create and solve one technical calculation involving triangulation. Show formula, substitution, units and interpretation.

5. Complex numbers

Rectangular and polar forms for advanced engineering mathematics.

In civil engineering, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
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- Interpret the result as a technician, not only as a number.

Main formula for this section

$$z=a+bi$$

Worked method

Example: Road reserve triangle. Given two sides and an included angle, use the cosine rule to find a missing boundary length.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports polar representation. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving slopes. Show formula, substitution, units and interpretation.

2. Create and solve one technical calculation involving triangulation. Show formula, substitution, units and interpretation.

3. Create and solve one technical calculation involving bearings. Show formula, substitution, units and interpretation.

4. Create and solve one technical calculation involving polar representation. Show formula, substitution, units and interpretation.

5. Create and solve one technical calculation involving heights. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving slopes. Show formula, substitution, units and interpretation.

7. Create and solve one technical calculation involving triangulation. Show formula, substitution, units and interpretation.

8. Create and solve one technical calculation involving bearings. Show formula, substitution, units and interpretation.

Mixed Revision Questions

1. A civil engineering problem involves slopes. Choose a suitable mathematical method, solve the problem, and explain the result.

2. A civil engineering problem involves triangulation. Choose a suitable mathematical method, solve the problem, and explain the result.

3. A civil engineering problem involves bearings. Choose a suitable mathematical method, solve the problem, and explain the result.

4. A civil engineering problem involves polar representation. Choose a suitable mathematical method, solve the problem, and explain the result.

5. A civil engineering problem involves heights. Choose a suitable mathematical method, solve the problem, and explain the result.

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9. A civil engineering problem involves polar representation. Choose a suitable mathematical method, solve the problem, and explain the result.

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23. A civil engineering problem involves bearings. Choose a suitable mathematical method, solve the problem, and explain the result.

24. A civil engineering problem involves polar representation. Choose a suitable mathematical method, solve the problem, and explain the result.

25. A civil engineering problem involves heights. Choose a suitable mathematical method, solve the problem, and explain the result.

26. A civil engineering problem involves slopes. Choose a suitable mathematical method, solve the problem, and explain the result.

27. A civil engineering problem involves triangulation. Choose a suitable mathematical method, solve the problem, and explain the result.

28. A civil engineering problem involves bearings. Choose a suitable mathematical method, solve the problem, and explain the result.

29. A civil engineering problem involves polar representation. Choose a suitable mathematical method, solve the problem, and explain the result.

30. A civil engineering problem involves heights. Choose a suitable mathematical method, solve the problem, and explain the result.

References and Learning Sources

These learner materials are original notes prepared for Mr Mwirigi Mathematics Training Hub. The topic sequencing and level are informed by standard engineering mathematics references and the uploaded curriculum/OS materials. No textbook pages or solution-manual pages have been reproduced.

1. John Bird, Engineering Mathematics, Newnes / Elsevier.

2. K. A. Stroud and Dexter J. Booth, Engineering Mathematics, Palgrave Macmillan.

3. Uploaded KNEC solved mathematics materials for examination-style orientation.

4. Civil Engineering, Building Technology and Land Survey curriculum and occupational-standard materials used for local alignment.